Scope of Midterm project

1. Time and mode.

Due for midterm project slides submission is 22:00 PM on 1 March 2026.
*You should submit the slides in **one** Pdf file to the **CANVAS** system.*

Presentation on 23, 25, 27 Feb. 2026.

2. Data and Methods (Choose one of the following options)

Option 1: Standard Dataset Analysis

You may select **two datasets** from either **Hugging Face** or the **UCI Machine Learning Repository**, and complete **either** a **classification** or **regression** task.

- Use the modeling techniques and evaluation criteria covered in **Lectures 1–21**.
- You may focus on **one task type only** (classification *or* regression).
- Ensure that all **seven required project components** are completed.

Scoring:

- **Baseline score:** 70 points (for completing all required contents)
- **Bonus points:** Awarded based on the overall quality, rigor, and completeness of the seven components

Option 2: Coupon Usage Prediction (Additional Bonus Opportunity)

As an advanced option, you may choose to improve classification performance using the **provided coupon usage dataset**.
Your objective is to **enhance classification accuracy** by effectively **merging three related datasets**:

- User consumption (wallet transaction data)
- Coupon issuance (coupon delivery data)
- Coupon redemption (coupon usage data)

The goal is to **predict the likelihood that users will respond to future marketing campaigns** based on their historical behavior and coupon-related information.
Please refer to the **attached data description** for detailed explanations of each dataset.

Scoring:

- **Baseline score:** 80 points (for completing all required contents)
- **Bonus points:** Awarded based on the quality, depth, and effectiveness of the seven components

Additional Performance Bonus:

- If your **testing accuracy exceeds 95%** and your code is **valid, reproducible, and well-documented**, you will receive an **additional 5 bonus points** on your total score for this course.

3. Project Contents

Your project report must include the following **seven components**, presented in order:

1) Task Definition

Clearly state the objective of your study.

- Separate **regression** and **classification** tasks into **distinct sections** if both are involved.
- Clearly define the prediction target and the problem setting.

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2) Data Description and Background

Describe the dataset(s) used in your project, including:

- The **response variable (target)** and **feature variables (predictors)**
- The data source and basic characteristics (e.g., size, variable types)
- Key patterns, trends, or relationships you aim to explore or verify

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3) Data Preprocessing

Explain all preprocessing steps applied to the data, such as:

- Data cleaning and handling of missing values
- Feature selection or feature engineering
- Data transformation, normalization, or encoding
- Train–test split or cross-validation setup

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4) Model Construction

Describe the modeling approach in detail:

- The methods and algorithms used
- Model assumptions (if applicable)

- Hyperparameter settings and model selection strategy
- Justification for choosing the model(s)

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5) Function Definition and Implementation

Define the key functions and libraries/packages used in **Python or R**:

- Clearly explain the purpose of each major function
- Display the **first five rows** of:
  - Input data
  - Intermediate outputs
  - Final results
    (e.g., using head() in R or .head() in pandas)

Include the relevant outputs (tables or summaries) directly in your report.

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6) Data Visualization

Present appropriate **figures and tables** to support your analysis, such as:

- Feature distributions
- Model performance metrics
- Comparison across models or settings

Ensure all visualizations are clearly labeled and easy to interpret.

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7) Data Analysis and Conclusion

Provide a brief written analysis that:

- Interprets the results and model performance
- Discusses key findings and insights
- Addresses limitations and possible improvements
- Summarizes conclusions and practical implications

4. Submission

For project report submission, where you should submit only the '.pdf' files of slides including the presentations.

5. Presentation

For the in-class presentation of your data analysis results, please prepare a concise and informative slideshow.

Begin with a brief introduction to your dataset and the objectives of your analysis. Then, outline the methodology and tools used. Highlight key

findings and insights, and conclude with implications or recommendations based on your results.
Aim for clarity and brevity, as each presentation should last approximately 5 minutes. Ensure your slides are visually engaging and your narrative is easy to follow.

6. Example
There are a lot of data analysis report samples in Kaggle. While exploring these templates for inspiration is encouraged, it's crucial to ensure that your work remains original. The university employs verification processes to check for direct content duplication.
The following example is from Kaggle: https://www.kaggle.com/code/bsivavenu/house-price-calculation-methods-for-beginners.